Data Analytics Project: Railway Guage Trends

1. Introduction:

This project focuses on analyzing railway gauge data using Python. The goal is to understand the structure of the dataset, clean the data, and visualize key trends related to railway track expansion. By using data analysis and visualization techniques, meaningful insights about railway development over time can be derived.

2. Objectives:

- To load railway gauge data from a CSV file
- To handle missing and inconsistent data
- To visualize total railway track growth over time
- To compare different railway gauges in recent years
- To identify trends and dominant gauge types

3. Technologies Used:

- Python
- Pandas(Data manipulation and analysis)
- Numpy(Numerical operations)
- Matplotlib(Data Visualization)

4. Scenario 1: Basic Data Loading & Cleaning

Problem Statement

Load the dataset, inspect its structure, handle missing values, and convert columns into appropriate data types for analysis.

Approach
1. Load the dataset into a Pandas DataFrame.
2. Display the first 5 rows and column names.
3. Check for missing values and replace them with 0.
4. Convert all gauge columns (Broad, Metre, Narrow, Total) to numeric types

Step 1: Load the dataset into a Pandas DataFrame.

Code

```
#1.1 Load the dataset

df=pd.read_csv('railway_gauges 1.csv')
```

Explanation:

The dataset is loaded from a CSV file using Pandas. This allows further data processing and analysis.

Step 2: Display the first 5 rows and column names.

Code

```
#1.2 Display the first 5 rows and column names
print("First 5 Rows: ")
print(df.head())

print("\nFirst 5 columns: ")
print(df.columns)
```

Explanation:

The first 5 rows are displayed to understand the structure of the dataset. Column names are printed to identify available fields.

Output

	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
0	1964-65	3521	2891	464	6876
1	1965-66	3603	2913	470	6986
2	1966-67	3615	2911	481	7007
3	1967-68	3601	2920	506	7027
4	1968-69	3603	2921	508	7032

Step 3: Check for missing values and replace them with 0.

Code

```
#1.3 Check for missing values and replace them with 0
print("\nMissing Values: ")
print(df.isnull().sum())
df=df.fillna(0)

print("\nAfter replacing with 0: ")
print(df.isnull().sum())
```

Explanation:

Missing values are identified and replaced with 0 to ensure clean data for analysis.

Output

```
Missing Values:
Year            0
Broad Gauge     0
Metre Gauge     0
Narrow Gauge    0
Total           0
dtype: int64

After replacing with 0:
Year            0
Broad Gauge     0
Metre Gauge     0
Narrow Gauge    0
Total           0
dtype: int64
```

Step 4: Convert all gauge columns (Broad, Metre, Narrow, Total) to numeric types.

Code:

```
#1.4 Convert all gauge columns (Broad, Metre, Narrow, Total) to numeric types

gauge_columns=["Broad Gauge","Metre Gauge","Narrow Gauge","Total"]

for col in gauge_columns:
    df[col]=pd.to_numeric(df[col])

print("\nData Types after conversion:")
print(df.dtypes)
```

Explanation:

Gauge columns are converted into numeric format to allow mathematical operations and visualization.

Output

```
Data Types after conversion:
Year                str
Broad Gauge       int64
Metre Gauge       int64
Narrow Gauge      int64
Total             int64
dtype: object
Total             int64
dtype: object
dtype: object
```

5. Scenario 2: Simple Visualization

Problem Statement

To visualize the growth of total railway track length over the years using a line graph and analyze whether the overall trend is increasing or decreasing.

Approach

1.Extract Year and Total columns.
2. Plot a line graph showing Total tracks over years.
3. Add: Title , X and Y labels
4. Identify whether the trend is increasing or decreasing.

Step 1: Extract Year and Total columns

Code

```
#2.1 Extract Year and Total columns

year=df['Year']
total=df["Total"]
```

Explanation:

The year and total columns are selected from the dataset to analyze the growth of railway track length over time using Pandas.

Step 2 & Step 3: Plot a line graph showing Total tracks over years, Add: Title , X and Y labels

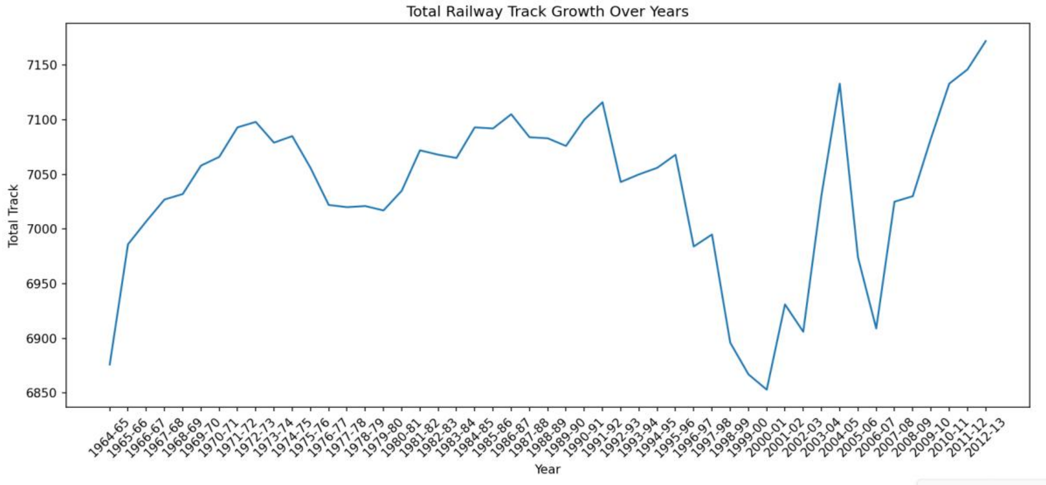
Code

```
#2.2 Plot a line graph showing Total tracks over years.
plt.figure()
plt.plot(year,total)

#2.3 Add: Title and X and Y labels

plt.title("Total Railway Track Growth Over Years")
plt.xlabel("Year")
plt.ylabel("Total Track")

plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("Railway_Gauges1S2.png")
plt.show()
```

Explanation:

A line graph is plotted using Matplotlib to visualize how total railway track length changes over the years, Title and axis labels are added to make the graph clear and understandable and X-axis labels are rotated and layout is adjusted to prevent overlapping and improve clarity.

Output



Step 4: Identify whether the trend is increasing or decreasing.

The line graph shows that the total railway track length increases over the years.
This indicates an overall increasing trend in railway infrastructure.

6. Scenario 3: Filtering + Bar Chart

Problem Statement

To analyze modern railway expansion by filtering the dataset for years after 2000 and comparing Broad Gauge, Metre Gauge, and Narrow Gauge using a grouped bar chart to identify the dominant gauge in recent years.

Approach

1. Filter the dataset for years after 2000.
2. Select Broad Gauge, Metre Gauge, and Narrow Gauge.
3. Plot a grouped bar chart comparing all three gauges.
4. Add legend and proper labels.
5. Identify which gauge dominates in recent years.

Step 1 Step 2:Filter the dataset for years after 2000, Select Broad Gauge, Metre Gauge, and Narrow Gauge.

Code:

```
#3.1 Filter the dataset for years after 2000.

df["Year"] = df["Year"].str.split('-').str[0]
df["Year"] = pd.to_numeric(df["Year"], errors='coerce')
df["Year"] = pd.to_numeric(df["Year"])
df_filtered=df[df["Year"]>2000]

#3.2 Select Broad Gauge, Metre Gauge, and Narrow Gauge.
years=df_filtered['Year']
broad_gauge=df_filtered["Broad Gauge"]
metre_gauge=df_filtered["Metre Gauge"]
narrow_gauge=df_filtered["Narrow Gauge"]
```

Explanation:

The dataset is filtered to include only records where the year is greater than 2000. This helps focus on modern railway expansion using Pandas.The Broad Gauge, Metre Gauge, and Narrow Gauge columns are selected for comparison.

Output:

	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
37	2001	5338	1310	283	6931
38	2002	5402	1220	284	6906
39	2003	5611	1190	230	7031
40	2004	5780	1134	219	7133
41	2005	5727	1018	229	6974
42	2006	5830	888	191	6909
43	2007	6000	806	219	7025
44	2008	6185	642	203	7030
45	2009	6320	543	220	7083
46	2010	6419	494	220	7133
47	2011	6494	432	220	7146
48	2012	6577	379	216	7172

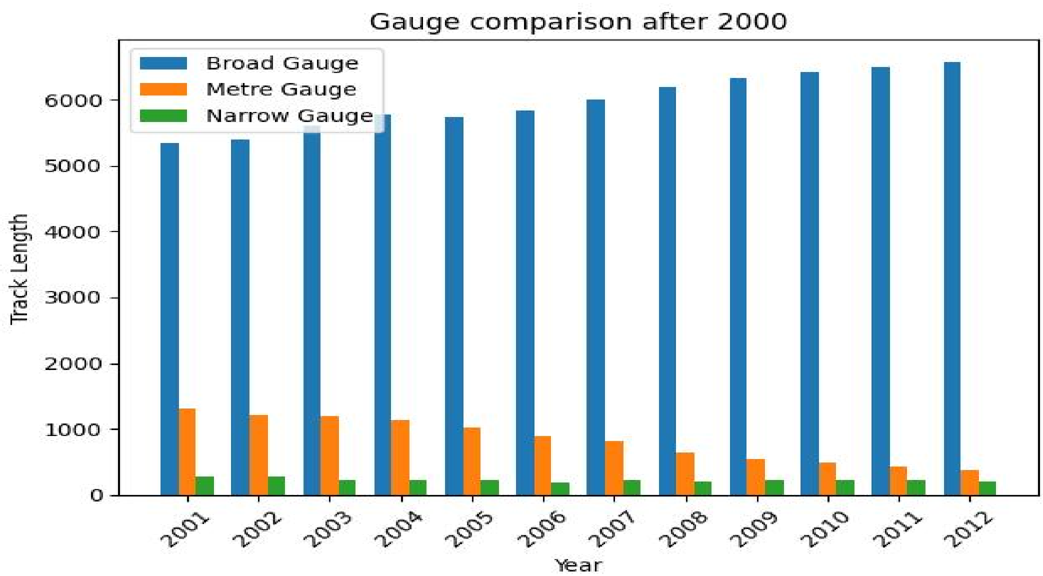
Step 3 & Step 4: Plot a grouped bar chart comparing all three gauges, Add legend and proper labels.

Code:

```
#3.3 Plot a grouped bar chart comparing all three gauges

x=np.arange(len(years))
width=0.25
plt.figure()
plt.bar(x - width,broad_gauge,width,label="Broad Gauge")
plt.bar(x,metre_gauge,width,label="Metre Gauge")
plt.bar(x + width,narrow_gauge,width,label="Narrow Gauge")

#3.4 Add legend and proper labels

plt.title("Gauge comparison after 2000")
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.xticks(x,years,rotation=45)
plt.legend()
plt.tight_layout()
# plt.savefig("Railway_Gauges2S3.png")
plt.show()
```

Explanation:
A grouped bar chart is created to compare the three gauge types for each year using Matplotlib and labels, title, and legend are added to improve clarity and readability of the chart.

Output:



Step 5:Identify which gauge dominates in recent years.

From the grouped bar chart, Broad Gauge consistently shows higher values compared to Metre and Narrow Gauge.

7. Scenario 4: Feature Engineering + Pie Chart

Problem Statement:

To analyze the contribution of each railway gauge type by calculating their total values and visualizing their percentage share using a pie chart.

Approach:

1. Calculate total sum of each gauge across all years.
2. Create a new structure (Series/DataFrame) for totals.
3. Plot a pie chart showing percentage contribution.
4. Add percentage labels (autopct).
5. Interpret which gauge contributes the most.

Step 1: Calculate total sum of each gauge across all years.

Code:

```
# 4.1 Calculate total sum of each gauge across all years.

total_broad=df["Broad Gauge"].sum()
total_metre=df["Metre Gauge"].sum()
total_narrow=df["Narrow Gauge"].sum()

print("\nTotal Broad Gauge:", total_broad)
print("Total Metre Gauge:", total_metre)
print("Total Narrow Gauge:", total_narrow)
```

Explanation:

The total values of Broad Gauge, Metre Gauge, and Narrow Gauge are calculated across all years using Pandas.

Output:

```
Total Broad Gauge: 224066
Total Metre Gauge: 101798
Total Narrow Gauge: 18953
```

Step 2: Create a new structure (Series/DataFrame) for totals.

Code:

```
#4.2 Create a new structure (Series/DataFrame) for totals.

gauge_totals=pd.Series([total_broad,total_metre,total_narrow],
                       index=["Broad Gauge","Metre Gauge","Narrow Gauge"])

print("\nGauge Totals:")
print(gauge_totals)
```

Explanation:

A new structure Series is created to store the total values of each gauge type.

Output:

```
Gauge Totals:
Broad Gauge     224066
Metre Gauge     101798
Narrow Gauge     18953
dtype: int64
```

Step 3 & Step 4: Plot a pie chart showing percentage contribution, Add percentage labels (autopct).

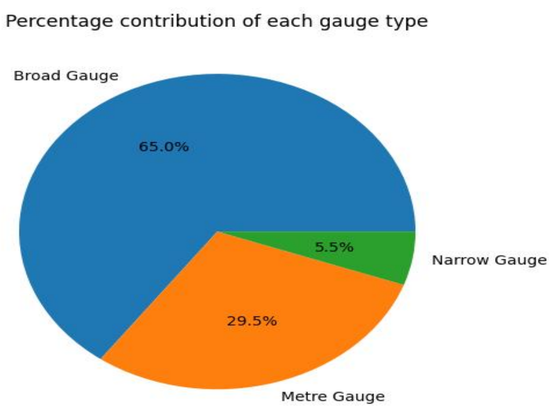
Code:

```
#4.3 & 4.4 Plot a pie chart showing percentage contribution,Add percentage labels (autopct).

plt.figure()
plt.pie(gauge_totals,labels=gauge_totals.index,autopct="%1.1f%%")
plt.title("Percentage contribution of each gauge type")
plt.tight_layout()
plt.savefig("Railway_Gauges3S4")
plt.show()
```

Explanation:

A pie chart is created using Matplotlib to visualize the contribution of each gauge type and Percentage labels are added to clearly show the contribution of each gauge type.

Output:



Step 5: Interpret which gauge contributes the most.

Broad Gauge contributes the highest percentage among all gauge types, indicating it plays the most significant role in the railway network.

8. Scenario 5:Advanced Analysis + Multiple Graphs

Problem Statement:

To perform an advanced analysis of railway gauge trends by creating percentage-based features, calculating yearly growth in total railway tracks, and visualizing trends using multiple graphs. The goal is to understand long-term railway development and identify whether a dominant gauge is emerging.

Approach:

1. Create new columns:
   - o % Broad Gauge
   - o % Metre Gauge
   - o % Narrow Gauge
2. Use NumPy (np.diff) to calculate yearly growth of Total tracks.
3. Plot:
   - o Line graph for all gaugeso Stacked bar chart showing composition
4. Highlight:
   - o Years with highest growth
   - o Decline in any gauge
5. Provide a final conclusion: "Is the railway system shifting towards a single dominant gauge?"

Step 1: Create new columns: ○ % Broad Gauge ○ % Metre Gauge ○ % Narrow Gauge

Code:

```
#5.1 Create new columns:o % Broad Gauge o % Metre Gauge o % Narrow Gauge

df["% Broad Gauge"] = (df["Broad Gauge"] / df["Total"]) * 100
df["% Metre Gauge"] = (df["Metre Gauge"] / df["Total"]) * 100
df["% Narrow Gauge"] = (df["Narrow Gauge"] / df["Total"]) * 100

print("\nPercentage Columns Added:")
print(df[["% Broad Gauge", "% Metre Gauge", "% Narrow Gauge"]].head())
```

Explanation:

New columns are created to represent the percentage contribution of Broad, Metre, and Narrow Gauge using Pandas.

Output:

```
Percentage Columns Added:
   % Broad Gauge  % Metre Gauge  % Narrow Gauge
0      51.207097      42.044793        6.748109
1      51.574578      41.697681        6.727741
2      51.591266      41.544170        6.864564
3      51.245197      41.554006        7.200797
4      51.237201      41.538680        7.224118
```

Step 2: Use NumPy (np.diff) to calculate yearly growth of Total tracks.

Code:

```
#5.2 Use NumPy (np.diff) to calculate yearly growth of Total tracks.

growth=np.diff(df["Total"])
growth = np.insert(growth, 0, 0)  # to match length

df["Yearly Growth"]=growth
print("\nYearly Growth:")
print(df[["Year", "Total", "Yearly Growth"]].head())
```

Explanation:

Yearly growth in total railway tracks is calculated using np.diff() from NumPy, which finds differences between consecutive years.

Output:

```
Yearly Growth:
   Year  Total  Yearly Growth
0  1964   6876              0
1  1965   6986            110
2  1966   7007             21
3  1967   7027             20
4  1968   7032              5
```

Step 3.1: Plot: ○ Line graph for all gauges

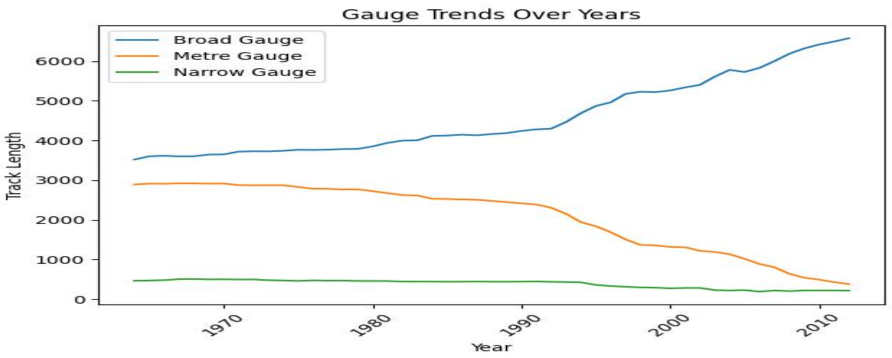
Code:

```
#5.3.1 plot Line graph for all gauges
plt.figure()
plt.plot(df["Year"],df["Broad Gauge"],label="Broad Gauge")
plt.plot(df["Year"],df["Metre Gauge"],label="Metre Gauge")
plt.plot(df["Year"],df["Narrow Gauge"],label="Narrow Gauge")

plt.title("Gauge Trends Over Years")
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("Railway_Gauges4S5.png")
plt.show()
```

Explanation:

A line graph is plotted to compare Broad, Metre, and Narrow Gauge trends over time using Matplotlib.

Output:



Step 3.2: ○ Stacked bar chart showing composition

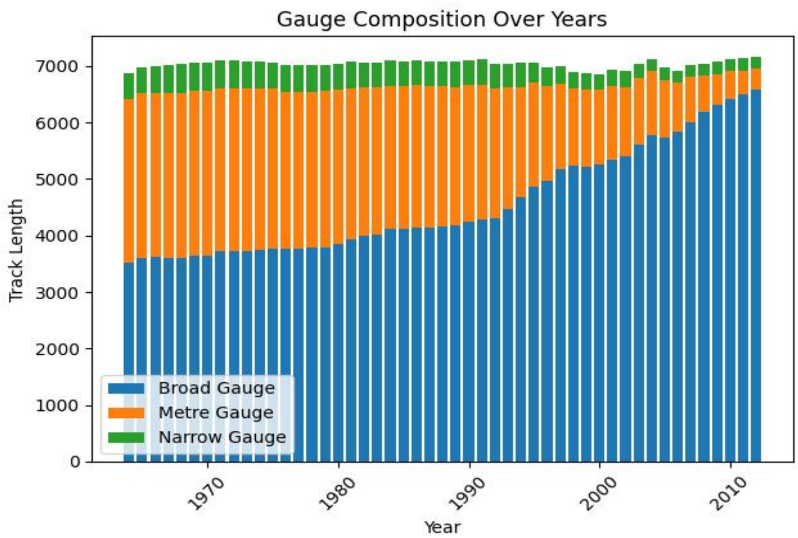
Code:

```
#5.3.2 Stacked bar chart showing composition

plt.figure()
plt.bar(df["Year"],df["Broad Gauge"],label="Broad Gauge")
plt.bar(df["Year"],df["Metre Gauge"],bottom=df["Broad Gauge"],label="Metre Gauge")
plt.bar(df["Year"],df["Narrow Gauge"],
        bottom=df["Broad Gauge"]+df["Metre Gauge"],label="Narrow Gauge")

plt.title("Gauge Composition Over Years")
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("Railway_Gauges5S5.png")
plt.show()
```

Explanation:

A stacked bar chart is used to show the composition of different gauge types in total railway tracks.

Output:



Step 4: Highlight:
○ Years with highest growth
○ Decline in any gauge

Code:

```
#5.4.1 Highlight: o Years with highest growth

max_growth_year = df.loc[df["Yearly Growth"].idxmax(), "Year"]
print("\nYear with Highest Growth:", max_growth_year)

#5.4.2 Identify Decline in any gauge

decline_broad = df[df["Broad Gauge"].diff() < 0]
decline_metre = df[df["Metre Gauge"].diff() < 0]
decline_narrow = df[df["Narrow Gauge"].diff() < 0]

print("\nBroad Gauge Decline Years:")
print(decline_broad["Year"])
print("\nMetre Gauge Decline Years:")
print(decline_metre["Year"])
print("\nNarrow Gauge Decline Years:")
print(decline_narrow["Year"])
```

Explanation:

The years with maximum growth in total railway tracks are identified using the growth values.Any decrease in gauge values across years is observed to identify declining trends.

The graphs show that Broad Gauge consistently dominates over time, while Metre and Narrow Gauge show comparatively lower contributions. Growth patterns indicate steady expansion of railway infrastructure with occasional fluctuations.

Step 5: Provide a final conclusion:

The analysis suggests that the railway system is gradually shifting towards a single dominant gauge (Broad Gauge), which plays the major role in modern railway expansion.

Final conclusion:

This project provided a comprehensive analysis of railway gauge data using Python. Across all scenarios, the dataset was cleaned, explored, and visualized to understand railway expansion trends. The analysis revealed that railway infrastructure has steadily increased over time, with Broad Gauge emerging as the dominant type. Visualization techniques helped in identifying patterns, comparisons, and growth trends effectively. Overall, the study highlights the importance of data analysis in understanding real-world transportation systems.